

PINNs for Chiral Topological Systems

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1 Introduction

2 1D Kitaev Chain

The one-dimensional (1D) Kitaev chain describes a p-wave superconductor constructed of N spinless fermions¹ characterised by the Bogoliubov-de Gennes (BdG) Hamiltonian,

$$H_{\text{Kitaev}} = -\mu \sum_n c_n^\dagger c_n + \sum_n \left[(-t c_{n+1}^\dagger c_n + \Delta c_n c_{n+1}) + \text{h.c.} \right] \quad (1)$$

where c_n and $c_n^\dagger$ are the spinless fermionic creation and annihilation operators, μ is the chemical potential, t is the hopping amplitude and Δ is the pairing amplitude. Tuning the chemical potential between $-4t \leq \mu \leq 4t$, initiates a topological phase transition as shown explicitly in Fig.??.

To briefly summarise, this transition occurs because fermionic operators can be written as super positions of a pair of Majorana operators. In the trivial regime, on-site Majorana operators are coupled together with even parity forming standard spinless fermions. In the non-trivial regime, the on-site Majorana operators decouple and couple to operators on adjacent sites either side. For a chain of finite dimension, two Majorana operators are left unpaired at the edges of the chain resulting in topological edge modes.[?]

As depicted in Fig.??, the transition occurs at $\mu/t \approx 2$ closing the superconducting gap. Proceeding this, bound states are fixed to zero-energy and a band gap, $\Delta_{\text{top}}(\mu)$, restored. These states manifest themselves as topological Majorana bound states at the edges of the Kitaev chain with increased exponential localisation for maximum $\Delta_{\text{top}}(\mu)$ at $\mu = 0$.

Although the Kitaev toy model differs from the model used in this paper, it clearly outlines the signatures of a topological phase transition. This example should act as a guide, highlighting key features to keep in mind whilst investigating the properties of the model described in Sec.2.

3 Physics-Informed Neural Networks

3.1 Architecture

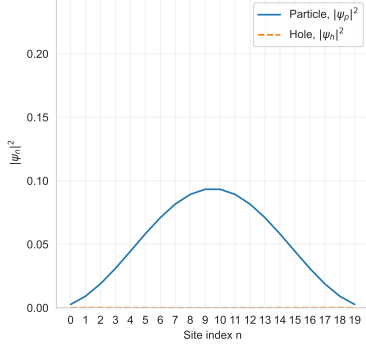
Dense neural network constituting of SIREN Layers

3.2 Optimisation

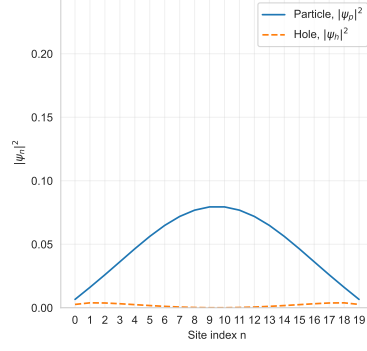
AdamW and L-BFGS

3.2.1 Data

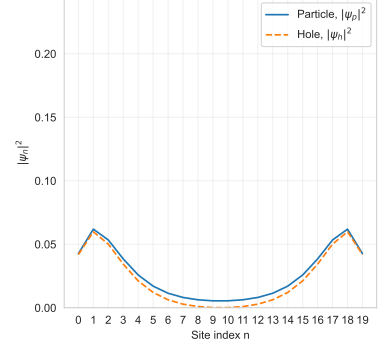
Diagonalisation for data loss and evaluation Collocation points for unsupervised - random sampling (fixed vs infinite sampled)



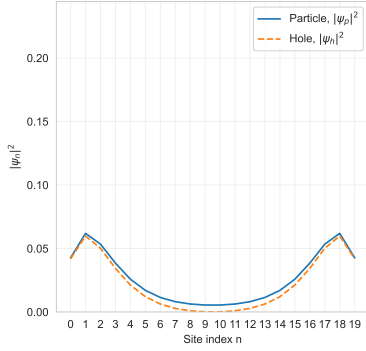
(a) $\mu = -4.0$



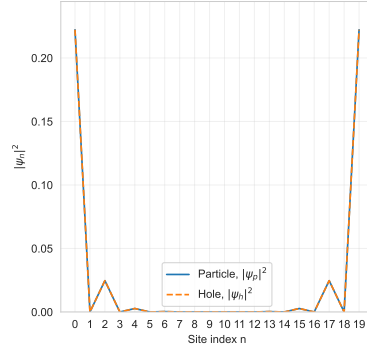
(b) $\mu = -2.2$



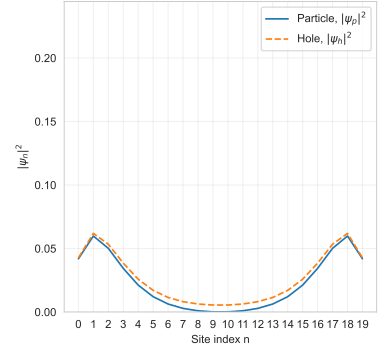
(c) $\mu = -2.0$



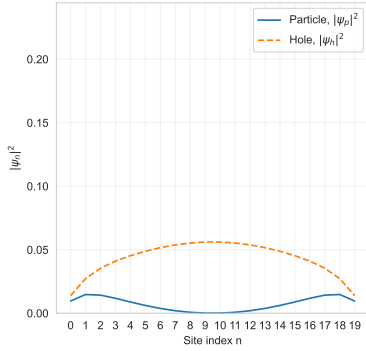
(d) $\mu = -1.8$



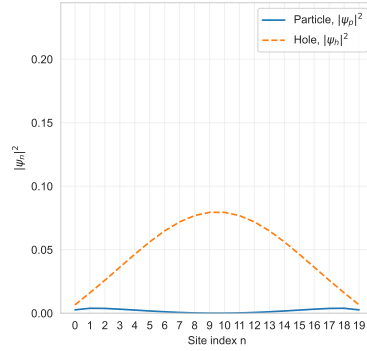
(e) $\mu = -0.0$



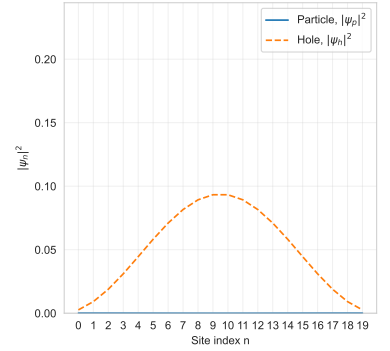
(f) $\mu = 1.8$



(g) $\mu = 2.0$



(h) $\mu = 2.2$



(i) $\mu = 4.0$

Figure 1: $|\psi_p|^2$ and $|\psi_h|^2$ for discrete μ between $-4t < \mu < 4t$.

3.2.2 Loss Function

- Data Loss - Physics Loss

3.2.3 Training

4 Methodology

4.1

5 1D Kitaev Chain

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where c_n and $c_n^\dagger$ are the spinless fermionic creation and annihilation operators, μ is the chemical potential, t is the hopping amplitude and Δ is the pairing amplitude. Tuning the chemical potential between $-4t \leq \mu \leq 4t$, initiates a topological phase transition as shown explicitly in Fig.??.

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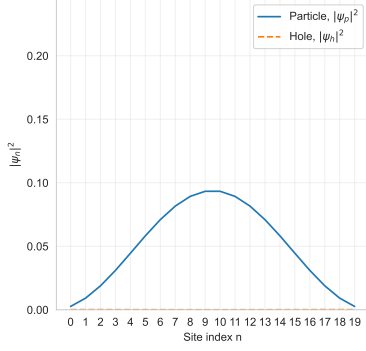
Although the Kitaev toy model differs from the model used in this paper, it clearly outlines the signatures of a topological phase transition. This example should act as a guide, highlighting key features to keep in mind whilst investigating the properties of the model described in Sec.2.

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N [y_i \log \sigma(f(x_i; \theta)) + (1 - y_i) \log(1 - \sigma(f(x_i; \theta)))] \quad (3)$$

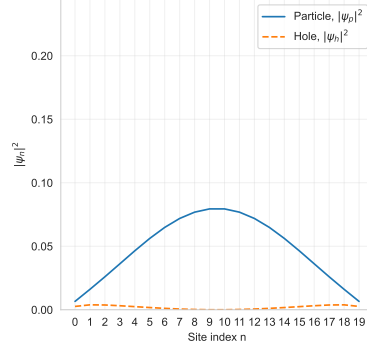
Equation 3 fits perfectly within the bounds of a single column.

6 Algorithms and Implementation

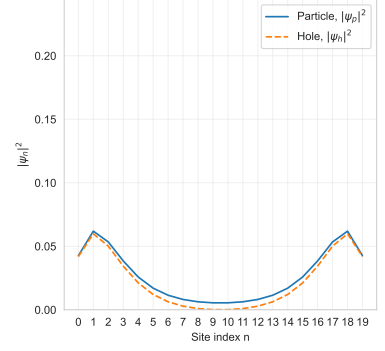
Pseudocode can be presented cleanly inside your column rules using the modern, non-conflicting `algpseudocode` syntax:



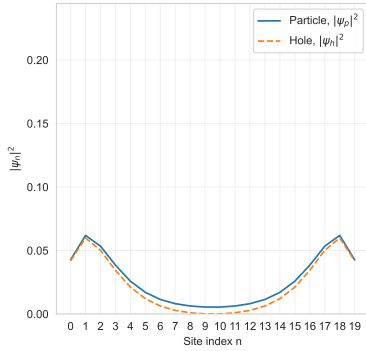
(a) $\mu = -4.0$



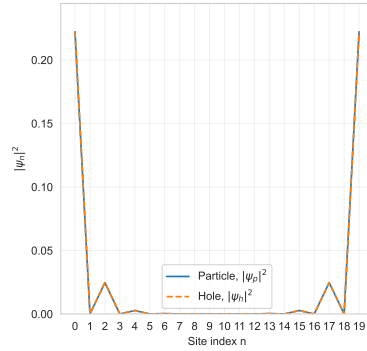
(b) $\mu = -2.2$



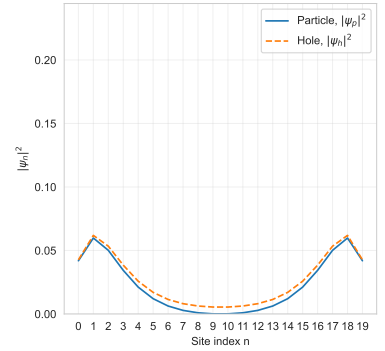
(c) $\mu = -2.0$



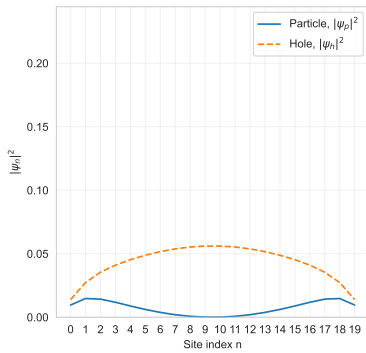
(d) $\mu = -1.8$



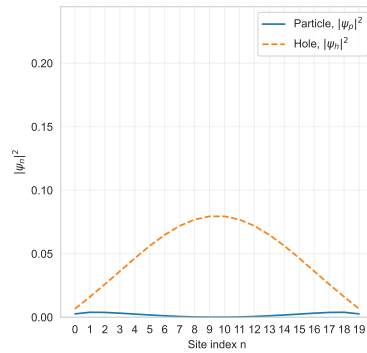
(e) $\mu = -0.0$



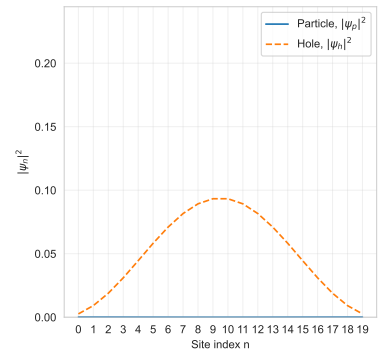
(f) $\mu = 1.8$



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(h) $\mu = 2.2$



(i) $\mu = 4.0$

Figure 2: $|\psi_p|^2$ and $|\psi_h|^2$ for discrete μ between $-4t < \mu < 4t$.

Algorithm 1 Stochastic Gradient Descent Update

Require: Learning rate η , initial parameters θ_0 , objective $\mathcal{L}$

Ensure: Optimized parameters θ

$\theta \leftarrow \theta_0$

while not converged **do**

 Sample mini-batch $\mathcal{B}$ from dataset

 Compute gradient $g \leftarrow \nabla_{\theta} \mathcal{L}(\mathcal{B}; \theta)$

$\theta \leftarrow \theta - \eta \cdot g$

end while

return θ

7 Results and Discussion

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7.1 Data and Parameters

Tables use clean horizontal rules from the `booktabs` package (`\toprule`, `\midrule`, and `\bottomrule`) instead of rigid vertical grid lines.

Table 1: Key hyperparameters used during our empirical evaluation.

Hyperparameter	Symbol	Value
Learning Rate	η	3×10^{-4}
Batch Size	B	256
Weight Decay	λ	1×10^{-5}

7.2 Visual Graphics

Standard figures scale directly to your individual column width. Set widths explicitly to `\columnwidth` to prevent layout bleeding:

If you ever need a massive image or table that must stretch horizontally across *both* columns, simply add an asterisk to the environment name, like this: `\begin{figure*}` or `\begin{table*}`.

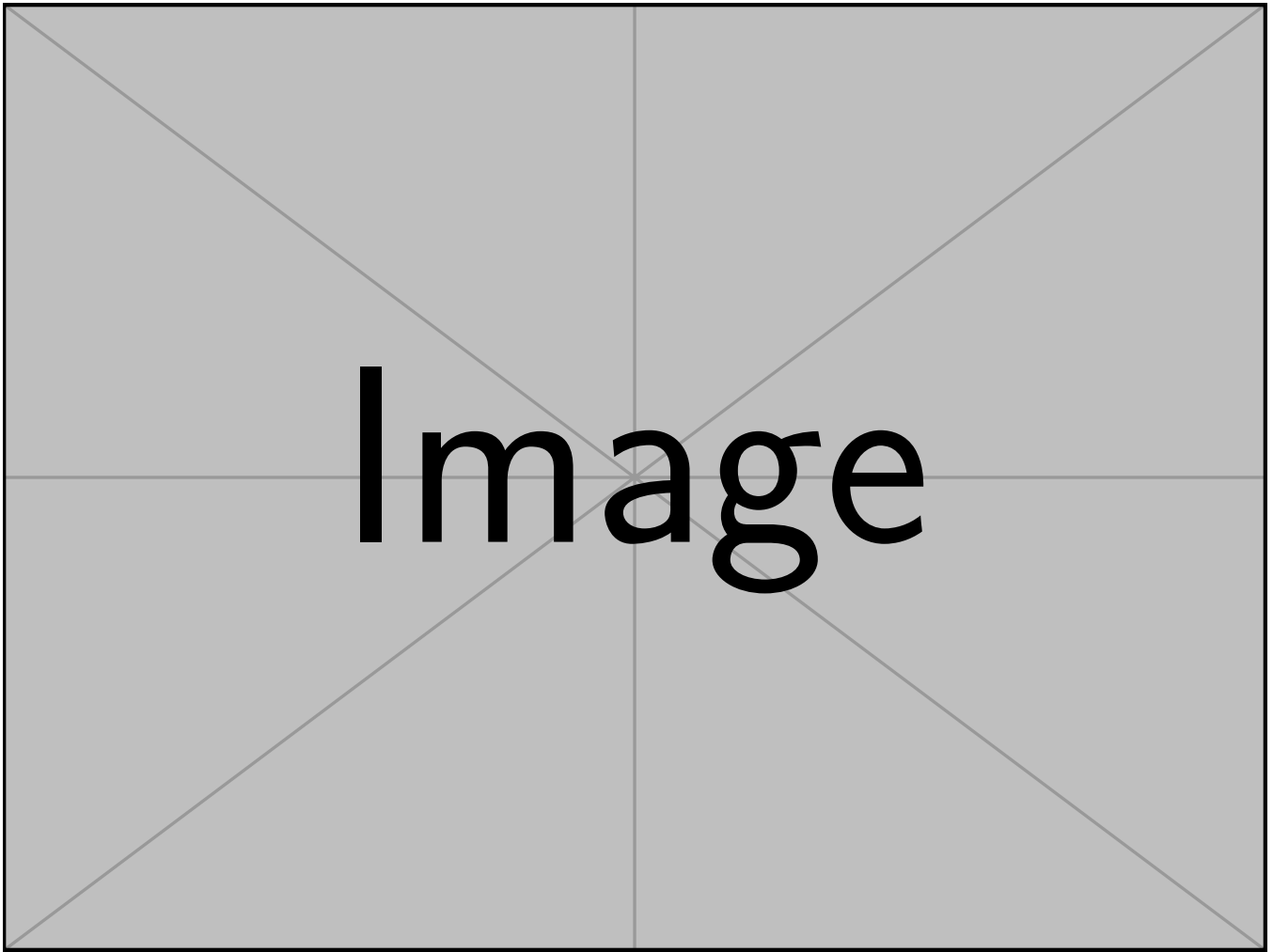


Figure 3: This is a center-aligned figure scaling uniformly within the bounds of a single text column.

8 Conclusion

This layout balances visual alignment with bulletproof compilation reliability. You can confidently expand this draft into a full publication manuscript.[?]

Acknowledgments

We thank our colleagues for their helpful commentary on this manuscript. This work was supported by the Template Design Foundation under Grant No. 000-12345.

References

¹ Alexei Yu. Kitaev. Unpaired majorana fermions in quantum wires. *Physics-Uspekhi*, 44(10S):131–136, 2001.